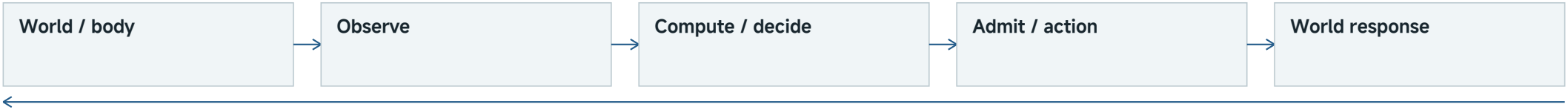
# 03 / System

What system produces that interaction?

**EX06 / NeuralFeels | Source analysis | C1 | P: paper; W: project; R: repository**




Feedback, operator enable/stop/recovery and evidence recording cross the loop.

## 1. World and observations

Draw embodiment/environment and sensing. Mark frames, measured quantities and implemented versus planned components.

An Allegro hand with DIGIT sensors and RGB-D sensing interacts with an object. [P 4.2]

## 2. Decision and action

Connect observations to computation/control and action. Declare action meaning, units, order, rates and limits.

A frontend estimates depth; neural-field and pose optimization fuse visual/tactile observations with kinematics. [P 4]

## 3. Authority and record

Add operator authority, logging, stop/reset/recovery and configuration ID. Identify evidence kept at each critical interface.

An existing rotation policy creates contact; calibration maps sensor measurements into shared coordinates. [P 4.2; S1]

## 4. Critical interface evidence

Select one consequential interface. State acceptance criterion, test evidence and result, or pending test with owner.

Next check: replay one contact and verify tactile-to-fingertip and hand-to-camera transforms. Not executed here. [R]

Review: If an interface is late, missing or misordered, how will you detect it and what action is allowed?

Example response: A wrong sensor transform can look like poor shape estimation.

# Worksheet 3 / System – evidence and team review

EX06 / NeuralFeels | SOURCE ANALYSIS EXAMPLE | Original review-page layout

Example ID: EX06  Project: NeuralFeels  Claim ID: C1  Revision: draft 1

Mode: Source analysis  Pages: canvas + this review; Decision also includes both A3 sides

Evidence scope: paper sections and public documentation; no dataset download or independent experiment.

| Evidence ID | Artifact/source + location (clickable) | What it supports; limitation / access gap |
|---|---|---|
| **P** | arxiv.org/abs/2312.13469v1 | Paper: method/results; reported evidence, not our measurement. |
| **W** | suddhu.github.io/neural-feels | Project: system/videos; selected examples do not establish reliability. |
| **R** | github.com/facebookresearch/neuralfeels | Repository: documented route; code and data not executed or audited. |

**Next action / owner / verification:**

A wrong sensor transform can look like poor shape estimation. Reviewer: inspect the cited basis before adapting this answer.

Teaching example: no actual student team or signatures. Source analysis is not a reproduced result; blank signature fields are intentional.

1. Name / student ID: _____

Signature: _____ Date: _____

2. Name / student ID: _____

Signature: _____ Date: _____

3. Name / student ID: _____

Signature: _____ Date: _____

4. Name / student ID: _____

Signature: _____ Date: _____

5. Name / student ID: _____

Signature: _____ Date: _____

6. Name / student ID: _____

Signature: _____ Date: _____

7. Name / student ID: _____

Signature: _____ Date: _____

8. Name / student ID: _____

Signature: _____ Date: _____

P/W/R identify linked public sources above. NE: not examined in this brief example, not evidence of failure. Support/context labels are provisional, not grades.